TECHNICAL SUPPLEMENT 08 · ENGLISH

Controller Specification, Measurements and Benchmark

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Controller and mission specification

The evaluated mission is A→B→A, A=(-1.00,31.06) m and B=(-0.70,-18.34) m in the Unity x-z plane. The reference one-way NavMesh path is 49.4009 m. Arrival radius is 2.0 m; robot-agent radius is 0.5 m; timeout is 180 s wall-clock. Completed paths are checked with the recorded start/end poses and waypoint regions, not a center-to-center bound.

Anisotropic magnitude C=exp(-longitudinal²/1.20²-lateral²/0.80²). Squared coordinates imply C(d,delta+pi)=C(d,delta). M4 selects the nearest eligible detected human within 4.0 m; target speeds are 0, 0.45, 0.6075 and 1.00 m/s across the intimate, personal, social and outside zones. There is no added M4 hysteresis state or sum over all human fields.

2. Measurement and quality interpretation

All methods use adjudicated endpoints in the supplied final table. Metadata retain 115 receiver-quality alerts and 383 UDP sequence gaps; sender-side endpoint adjudication is documented in the supplied archive. One zero-duration identical-pose interval is retained as an explicit time-quality case. Raw trajectories and telemetry are excluded here. Tables and analyses can be recomputed from the supplied adjudicated per-run metrics, not all metrics from raw streams.

The nominal speed target is not a strict bound on realized pose-derived speed. The observed B4 maximum is 1.215735 m/s and its filtered maximum is 1.101147 m/s; these values are retained, not clipped. Native NavMesh and manual actuation differ, limiting attribution of every motion difference solely to local-policy mathematics.

3. Separate computational benchmark

Method	n	Kernel mean ms	Mean run p95 ms	Command API ms	Misses
M0	20	0.0237	0.0348	0.0596	0
M1	20	0.0124	0.0251	0.0515	0
M2	20	0.0063	0.0214	0.0439	0
M3	20	0.0244	0.0384	0.0592	0
M4	20	0.0211	0.0293	0.0527	0
B1	20	17.3713	27.1229	17.4328	1
B2	20	0.0691	0.1153	0.1039	0
B3	20	0.0463	0.0716	0.0796	0
B4	20	1.8608	2.9150	1.9012	0

180 benchmark assignments = nine methods × E1/E4 × ten repetitions. The common control period is 0.10 s (10 Hz); the first five warm-up invocations are excluded from the recorded timing summaries. Kernel instrumentation and sensor-to-command API measurements do not measure physical sensor-to-actuator latency or certify hard real-time deadlines.

Session	Samples	CPU one-core %	CPU total %	RSS MB
20261019_112314	28100	26.823	1.676	248.533
20261019_203708	18068	28.855	1.803	248.164

CPU summaries concern the receiver process and its sessions, not comparable per-controller Unity CPU loads.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.